Do Income Groups Differ in Financial Structure? A Multivariate Analysis of Variance

1 Motivation

The main report used linear discriminant analysis to *classify* economies into income groups, achieving high cross-validated accuracy. Classification accuracy, however, is not a hypothesis test: it does not tell us whether the income groups differ in their underlying multivariate financial structure beyond what chance would produce, nor *which* dimensions drive any difference. Multivariate analysis of variance (MANOVA) supplies exactly that inferential complement. It tests the null hypothesis that the four income groups share a common mean vector across the financial indicators, and—through follow-up tests—localises the difference to specific components and indicators.

2 Methods

To avoid the singular covariance matrix produced by the near-collinear raw indicators (documented in the factor-analysis addendum: one perfect duplicate and seven further pairs with $|r| > 0.9$), MANOVA is run on the 7 principal-component scores with eigenvalue > 1—the same orthogonal inputs used for the LDA. The model is a one-way MANOVA with World Bank income group (four levels) as the factor.

Four multivariate test statistics are reported (Pillai's trace, Wilks' lambda, the Hotelling–Lawley trace, and Roy's largest root). Because the groups are of unequal size (Low income $n = 11$) and the equal-covariance assumption is suspect, **Box's M test** is computed; where it rejects homogeneity, Pillai's trace—the statistic most robust to covariance heterogeneity—is given primacy. Significant omnibus results are decomposed in three ways: univariate ANOVAs on each principal component (which axes differ), pairwise group MANOVAs with Bonferroni correction (which groups differ), and univariate ANOVAs on one representative raw indicator per theoretical dimension (which financial features differ, and in which direction).

3 Results

3.1 Omnibus test

All four statistics reject the null decisively (Pillai's $V = 0.756$, approx. $F(21, 309) = 4.95$, $p < 10^{-10}$; Wilks' $\Lambda = 0.353$, $p < 10^{-13}$). The income groups occupy genuinely different regions of financial-structure space; the effect is large (Pillai's $V = 0.76$, equivalently $1 - \Lambda = 0.65$ of the multivariate variance associated with group membership).

Box's M test strongly rejects homogeneity of the group covariance matrices ($M = 304$, $\chi^2(84) = 259$, $p < 10^{-15}$). The equal-covariance assumption therefore fails—consistent with the main report's finding that QDA slightly outperforms LDA—so inference rests on Pillai's trace, which remains

Table 1: One-way MANOVA of income group on the seven retained PC scores. All four statistics reject the null of equal group mean vectors.

Test	Statistic	Approx. F	df1	df2	p
Pillai	0.756	4.95	21	309	6.3e-11
Wilks	0.353	6.05	21	290.568	8.0e-14
Hotelling-Lawley	1.538	7.30	21	299	< 1e-16
Roy	1.331	19.58	7	103	1.9e-16

significant by a wide margin. (Box's M is itself sensitive to non-normality and to the small Low-income group, so it is read as corroborating, not decisive.)

3.2 Which components differ

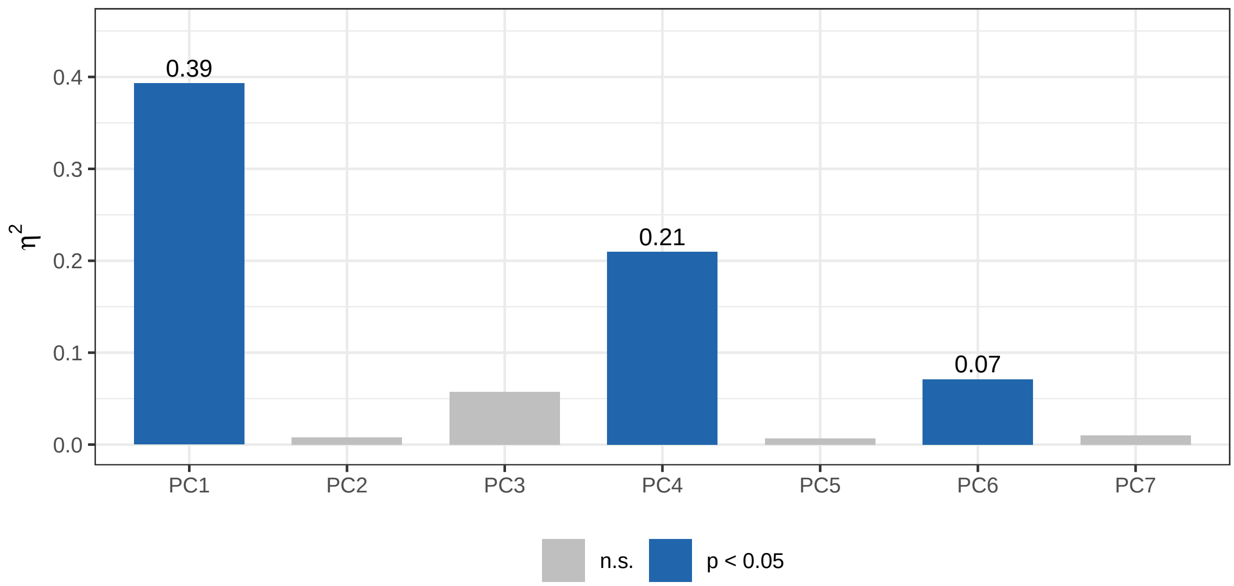

Figure 1: Univariate ANOVA effect sizes (η^2) for each principal component. Bars are shaded where the component differs significantly across income groups ($p < 0.05$). Group separation is concentrated on PC1 (financial depth and access) and PC4; the efficiency-contrast axis PC2 does not differ.

The omnibus difference is driven almost entirely by **PC1** ($\eta^2 = 0.39$, $p < 10^{-10}$)—the financial-depth-and-access axis—and by **PC4** ($\eta^2 = 0.21$, $p < 10^{-4}$). The remaining components, including the efficiency-contrast axis PC2, do not differ significantly across income groups (Figure 1).

3.3 Which groups differ

High-income economies differ significantly from every other group (all Bonferroni $p < 10^{-3}$). Among the lower groups, however, the picture is a continuum: Upper-Middle vs. Lower-Middle ($p_{\text{Bonf.}} = 0.08$) and Lower-Middle vs. Low ($p_{\text{Bonf.}} = 0.54$) are *not* distinguishable after correction. The clear multivariate boundary is therefore "High income vs. the rest," with adjacent lower groups

Table 2: Pairwise MANOVA (Wilks' lambda) between income groups, with Bonferroni correction for six comparisons.

Group A	Group B	p	p (Bonf.)
High	Upper-Mid.	8.7e-06	5.2e-05
High	Lower-Mid.	1.5e-10	8.8e-10
High	Low	4.5e-10	2.7e-09
Upper-Mid.	Lower-Mid.	0.013	0.07540
Upper-Mid.	Low	1.9e-05	0.00012
Lower-Mid.	Low	0.090	0.53997

blurring into one another—mirroring the adjacent-group misclassifications in the LDA confusion matrix.

3.4 Which financial features differ

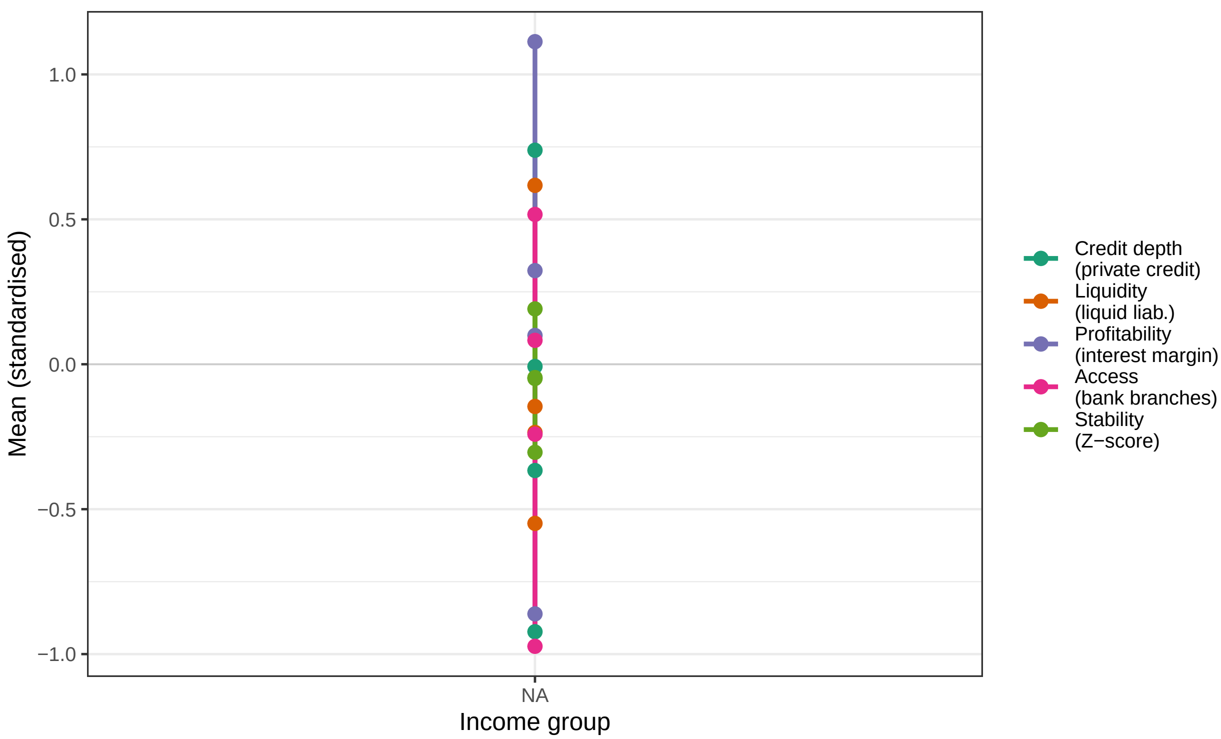

Figure 2: Standardised group means (z-scores) across income groups for one representative indicator per theoretical dimension. Credit, liquidity and access decline monotonically with income; the net interest margin is reversed (highest in low-income economies); stability (Z-score) is essentially flat.

The representative-indicator profiles (Figure 2, Table 3) make the substantive story concrete. **Credit depth, liquidity and access** all decline monotonically from High- to Low-income economies (all $p < 10^{-3}$), the expected gradient. **Profitability**, proxied by the net interest margin, runs in the *opposite* direction—margins are widest in Low-income economies and narrowest in High-income ones ($p < 10^{-10}$)—reflecting less competitive, higher-spread banking systems at lower income levels.

Table 3: Univariate ANOVA by income group for one representative indicator per dimension.

Dimension	F	p
Credit depth (private credit)	13.90	1.0e-07
Liquidity (liquid liab.)	6.86	0.00028
Profitability (interest margin)	20.80	1.1e-10
Access (bank branches)	8.37	4.7e-05
Stability (Z-score)	0.80	0.49815

Stability, proxied by the bank Z-score, does *not* differ across income groups ($F = 0.80$, $p = 0.50$), reinforcing the factor-analysis finding that the stability indicators do not behave as a coherent, income-linked construct.

4 Conclusions

MANOVA converts the descriptive separation seen in PCA and LDA into a formal result: income groups differ in financial structure far beyond chance (Pillai's $V = 0.76$, $p < 10^{-10}$), with a large effect size. The difference is not uniform. It is carried by **financial depth and access** (PC1) and is essentially a **"High income vs. the rest"** contrast—adjacent lower-income groups are statistically indistinguishable, confirming that income maps onto a financial-development *continuum* rather than four discrete regimes. Substantively, depth, liquidity and access rise with income while interest margins fall, and **stability shows no income gradient at all**. Because Box's M rejects covariance homogeneity, these conclusions are anchored on Pillai's trace and are consistent with the preference for quadratic over linear discriminant analysis noted in the main report.

5 Appendix: R Code

```
knitr::opts_chunk$set(
  echo    = FALSE,
  message = FALSE,
  warning = FALSE,
  fig.align = "center"
)
library(tidyverse)
library(readxl)
library(knitr)
library(kableExtra)

tmp <- tempfile(fileext = ".xlsx")
invisible(file.copy("Proposal/20220909-global-financial-development-database.csv", tmp))
raw <- read_excel(tmp, sheet = "Data - August 2022")
vars23 <- c(
  "di01","di02","di05","di08","di14",          # Depth (5)
  "ai02","ai25",                                 # Access (2)
  "ei01","ei03","ei04","ei05","ei06",
  "ei07","ei08","ei09","ei10",                   # Efficiency (9)
  "si01","si04","si06",                          # Stability (3)
  "oi01","oi02","oi13","oi14"                    # Other (4)
)

df2019 <- raw |>
  filter(year == 2019) |>
  dplyr::select(iso3, country, region, income, all_of(vars23)) |>
  drop_na() |>
  mutate(income = factor(income,
    levels = c("High income", "Upper middle income",
               "Lower middle income", "Low income")))

X      <- scale(as.matrix(df2019[, vars23]))
income <- df2019$income

inc_labels <- c("High income" = "High", "Upper middle income" = "Upper-Mid.",
                "Lower middle income" = "Lower-Mid.", "Low income" = "Low")
inc_colors <- c("High income" = "#2166ac", "Upper middle income" = "#92c5de",
                "Lower middle income" = "#f4a582", "Low income" = "#d6604d")

# PCA scores (Kaiser criterion: eigenvalue > 1) -- same input as the LDA
pca  <- prcomp(X, scale. = FALSE)
n_pc <- sum(pca$sdev^2 > 1)
Y    <- pca$x[, 1:n_pc]
fit <- manova(Y ~ income)
```

```r
mv_tab <- map_dfr(c("Pillai","Wilks","Hotelling-Lawley","Roy"), function(tst) {
  s <- summary(fit, test = tst)$stats
  tibble(Test = tst, Statistic = s[1, 2], `Approx. F` = s[1, 3],
         `df1` = s[1, 4], `df2` = s[1, 5], `p` = s[1, 6])
})
wilks  <- summary(fit, test = "Wilks")$stats[1, 2]
pillai <- summary(fit, test = "Pillai")$stats[1, 2]
# Box's M test for homogeneity of covariance matrices (manual implementation).
boxM <- function(Y, grp) {
  grp <- factor(grp); p <- ncol(Y); g <- nlevels(grp); N <- nrow(Y)
  ni <- table(grp); Si <- by(as.data.frame(Y), grp, cov)
  Sp <- Reduce(`+`, Map(function(S, n) (n - 1) * S, Si, ni)) / (N - g)
  M  <- (N - g) * log(det(Sp)) -
        sum((ni - 1) * sapply(Si, function(S) log(det(S))))
  c1 <- (sum(1/(ni - 1)) - 1/(N - g)) *
        (2*p^2 + 3*p - 1) / (6*(p + 1)*(g - 1))
  X2 <- M * (1 - c1); df <- p*(p + 1)*(g - 1)/2
  list(M = M, chisq = X2, df = df, p = pchisq(X2, df, lower.tail = FALSE))
}
bm <- boxM(Y, income)
aovs <- summary.aov(fit)
uni_tab <- map_dfr(seq_len(n_pc), function(i) {
  tab <- aovs[[i]]
  tibble(PC = paste0("PC", i),
         Fval = tab[1, "F value"],
         p    = tab[1, "Pr(>F)"],
         eta2 = tab[1, "Sum Sq"] / sum(tab[, "Sum Sq"]))
})
pairs <- combn(levels(income), 2)
pw_tab <- map_dfr(seq_len(ncol(pairs)), function(j) {
  idx <- income %in% pairs[, j]
  pw  <- manova(Y[idx, ] ~ droplevels(income[idx]))
  pv  <- summary(pw, test = "Wilks")$stats[1, 6]
  tibble(GroupA = inc_labels[pairs[1, j]],
         GroupB = inc_labels[pairs[2, j]],
         p = pv, p_bonf = pmin(pv * 6, 1))
})
reps <- c(di01 = "Credit depth\n(private credit)",
          di05 = "Liquidity\n(liquid liab.)",
          ei01 = "Profitability\n(interest margin)",
          ai02 = "Access\n(bank branches)",
          si01 = "Stability\n(Z-score)")
Xdf <- as.data.frame(X); Xdf$income <- income

rep_means <- map_dfr(names(reps), function(v) {
  m  <- tapply(Xdf[[v]], income, mean)
  av <- summary(aov(Xdf[[v]] ~ income))[[1]]
```

```r
  tibble(Dimension = reps[v], income = names(m), mean = as.numeric(m),
         F = av[1, "F value"], p = av[1, "Pr(>F)"])
}) |>
  mutate(income = factor(income, levels = levels(income)),
         Dimension = factor(Dimension, levels = reps))

rep_test <- rep_means |> distinct(Dimension, F, p)

# Scalars for inline reporting (avoids backtick-quoted names in inline code)
eta_pc1  <- uni_tab$eta2[1]
eta_pc4  <- uni_tab$eta2[4]
p_umlm   <- pw_tab$p_bonf[4]
p_lmlow  <- pw_tab$p_bonf[6]
stab_row <- rep_test[rep_test$Dimension == reps["si01"], ]
F_stab   <- stab_row$F
p_stab   <- stab_row$p
mv_print <- mv_tab |>
  mutate(Statistic = sprintf("%.3f", Statistic),
         `Approx. F` = sprintf("%.2f", `Approx. F`),
         df1 = sprintf("%g", df1), df2 = sprintf("%g", df2),
         p = format.pval(p, digits = 2, eps = 1e-16))
kable(mv_print, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "One-way MANOVA of income group on the seven retained PC scores. All four stat
      linesep = "") |>
  kable_styling(latex_options = "hold_position", font_size = 10)
uni_tab |>
  mutate(sig = p < 0.05,
         PC = factor(PC, levels = paste0("PC", 1:n_pc))) |>
  ggplot(aes(PC, eta2, fill = sig)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = ifelse(sig, sprintf("%.2f", eta2), "")),
            vjust = -0.4, size = 3) +
  scale_fill_manual(values = c(`TRUE` = "#2166ac", `FALSE` = "grey75"),
                    labels = c(`TRUE` = "p < 0.05", `FALSE` = "n.s."),
                    name = NULL) +
  labs(x = NULL, y = expression(eta^2)) +
  ylim(0, max(uni_tab$eta2) * 1.15) +
  theme_bw(base_size = 10) +
  theme(legend.position = "bottom")
pw_print <- pw_tab |>
  mutate(p = format.pval(p, digits = 2, eps = 1e-16),
         p_bonf = format.pval(p_bonf, digits = 2, eps = 1e-16))
kable(pw_print, format = "latex", booktabs = TRUE, row.names = FALSE,
      col.names = c("Group A", "Group B", "p", "p (Bonf.)"),
      caption = "Pairwise MANOVA (Wilks' lambda) between income groups, with Bonferroni correc
      linesep = "") |>
  kable_styling(latex_options = "hold_position", font_size = 10)
```

```r
ggplot(rep_means, aes(income, mean, colour = Dimension, group = Dimension)) +
  geom_hline(yintercept = 0, colour = "grey80", linewidth = 0.3) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2.3) +
  scale_x_discrete(labels = inc_labels) +
  scale_colour_brewer(palette = "Dark2") +
  labs(x = "Income group", y = "Mean (standardised)", colour = NULL) +
  theme_bw(base_size = 10) +
  theme(legend.position = "right")
sub_print <- rep_test |>
  mutate(Dimension = gsub("\n", " ", Dimension),
         F = sprintf("%.2f", F),
         p = format.pval(p, digits = 2, eps = 1e-16))
kable(sub_print, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "Univariate ANOVA by income group for one representative indicator per dimensi
      linesep = "") |>
  kable_styling(latex_options = "hold_position", font_size = 10)
```